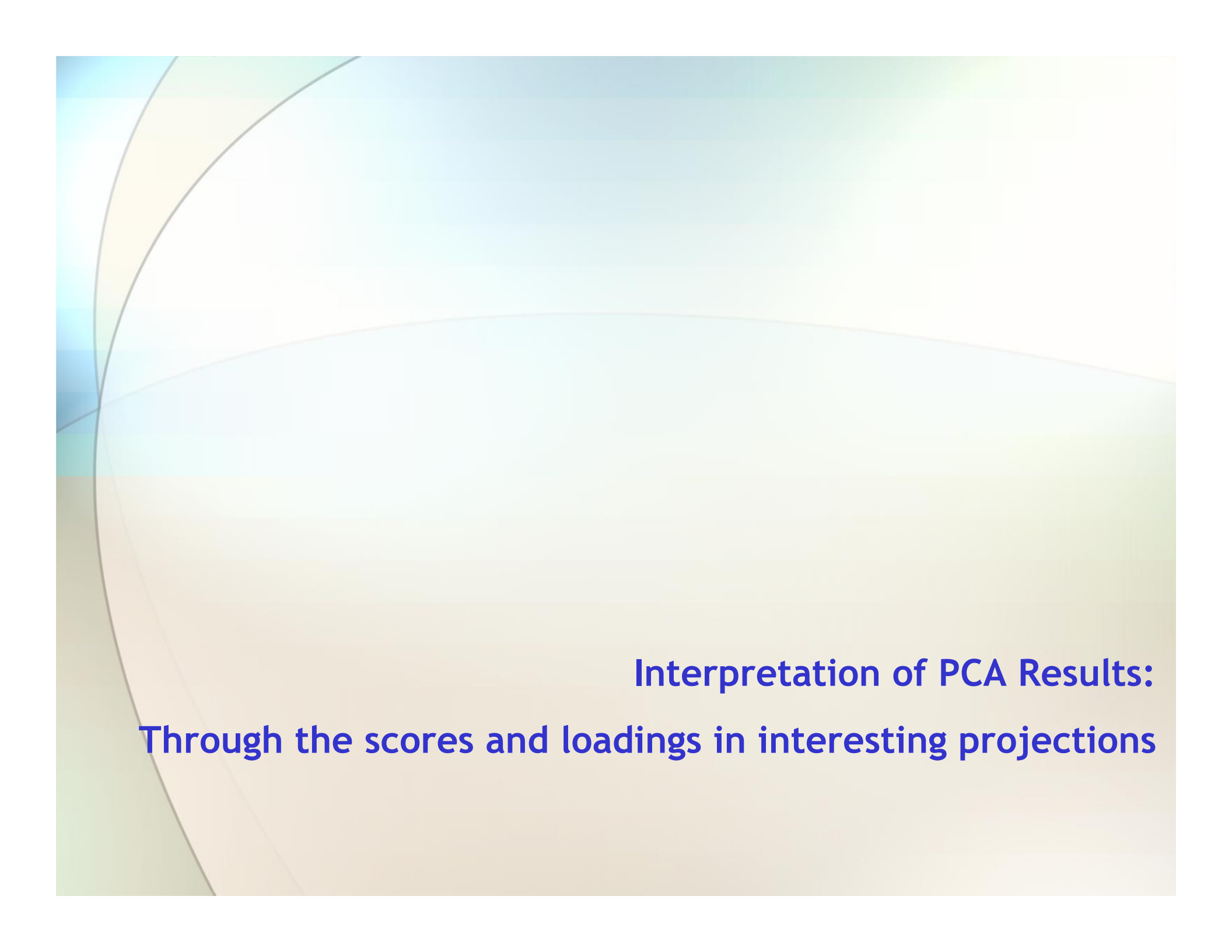
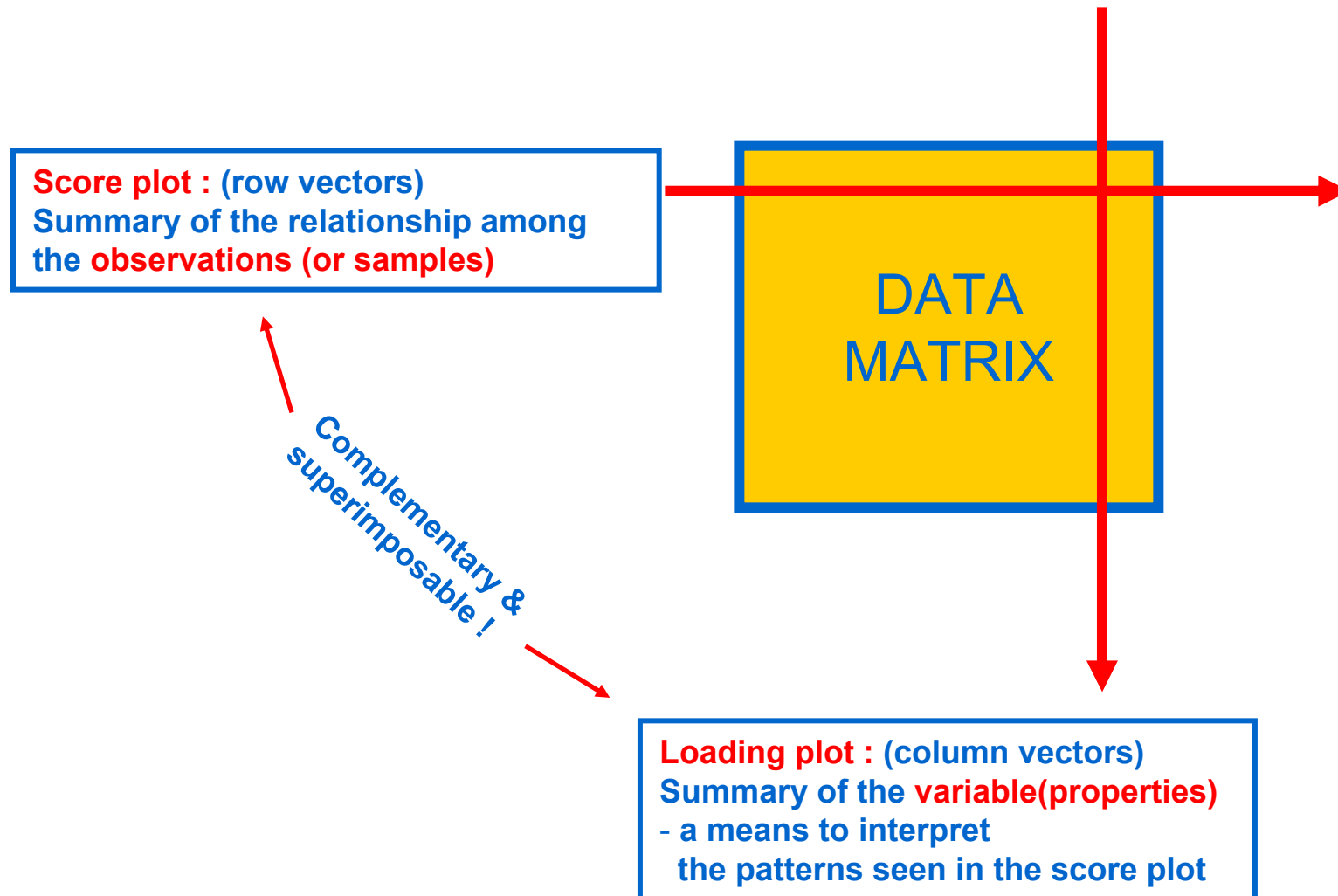




Interpretation of PCA Results:
Through the scores and loadings in interesting projections

Interpretation of scores and loadings



Example: Interpretation of scores and loadings with semiconductor data

A=

compounds	atomic no.	melting point	VE	radii	EN	lattice const.(ang)	compounds	atomic no.	melting point	VE	radii	EN	lattice const.(ang)
AlN	10	498	4	1.135	-1.21	3.11	(ZnMg) _{0.5} S	18.5	597.5	6.5	0.78	-1.21	5.52
AlP	14	625	4	0.435	-0.68	5.47	(SSe) _{0.5} Mg	18.5	681.5	4	-0.93	1.34	5.81
AlAs	23	1011.5	4	0.26	-0.63	5.66	(SSe) _{0.5} Zn	27.5	566.5	9	-0.78	1.21	5.52
AlSb	32	918.5	4	-0.09	-0.5	6.14	(ZnMg) _{0.5} Se	27.5	650.5	6.5	0.595	-1.1	5.78
GaN	19	182.5	4	1.155	-1.15	3.16	(ZnCd) _{0.5} Se	36.5	568.5	9	0.595	-1.1	5.86
GaP	23	309.5	4	0.455	-0.62	5.45	(SeTe) _{0.5} Zn	36.5	650	9	-0.595	1.1	5.9
GaAs	32	545	2.5	0.28	-0.57	5.65	(SeTe) _{0.5} Cd	45.5	601	9	-0.93	1.14	6.28
GaSb	41	603	4	-0.07	-0.44	6.1	(ZnCd) _{0.5} Te	45.5	682.5	9	0.21	-0.94	6.27
InN	28	246	4	1.51	-1.22	3.54	(AlGa) _{0.5} P	18.5	467.25	4	0.435	-0.68	5.46
InP	32	373	4	0.81	-0.69	5.87	(PAs) _{0.5} Ga	27.5	502.75	4	-0.455	0.62	5.61
InAs	41	759.5	4	0.635	-0.64	6.06	(AlGa) _{0.5} As	27.5	853.75	4	0.26	-0.63	5.65
InSb	50	666.5	4	0.285	-0.51	6.48	(GaIn) _{0.5} P	27.5	341.25	4	0.455	-0.62	5.68
ZnS	23	540	9	0.78	-1.21	5.41	(AlIn) _{0.5} P	23	499	4	0.435	-0.68	5.69
ZnSe	32	593	9	0.595	-1.1	5.67	(AsSb) _{0.5} Al	27.5	965	4	-0.26	0.63	5.88
ZnTe	41	707	9	0.21	-0.94	6.1	(AlZn) _{0.5} As	27.25	951.25	6.25	0.26	-0.63	5.87
CdSe	41	544	9	0.93	-1.14	6.05	(AsSb) _{0.5} Ga	36.5	649.5	4	-0.28	0.57	5.85
CdTe	50	658	9	0.545	-0.98	6.48	(AlGa) _{0.5} Sb	36.5	760.75	4	-0.09	-0.5	6.11
MgS	14	655	4	0.93	-1.34	5.7	(GaIn) _{0.5} As	36.5	727.75	4	0.28	-0.57	5.82
MgSe	23	708	4	0.745	-1.23	5.9	(PAs) _{0.5} In	36.5	566.25	4	-0.81	0.69	5.94
(AlGa) _{0.5} N	14.5	340.25	4	1.135	-1.21	3.14	(AlIn) _{0.5} Sb	41	792.5	4	-0.09	-0.5	6.27
(AlIn) _{0.5} N	19	372	4	1.135	-1.21	3.32	(GaIn) _{0.5} Sb	45.5	634.75	4	-0.07	-0.44	6.26
(GaIn) _{0.5} N	23.5	214.25	4	1.155	-1.15	3.31	(AsSb) _{0.5} In	45.5	713	4	-0.635	0.64	6.24

Size of this data matrix, **A**, is 44×6 from 44 samples with 6 variables.

Example: Interpretation of scores and loadings with semiconductor data (Continued)

1. Scaling: **auto scaling** is used because variables have different units.

$$X = \text{auto-scaled matrix of } A = \begin{bmatrix} -1.9425 & -0.5401 & -0.5887 & 1.3495 & -0.9224 & -2.5417 \\ -1.5621 & 0.1188 & -0.5887 & 0.2139 & -0.2332 & -0.0565 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1.4335 & 0.1694 & -0.5887 & -0.6054 & 0.0789 & 0.7754 \\ 1.4335 & 0.5754 & -0.5887 & -1.5220 & 1.4833 & 0.7544 \end{bmatrix} \left. \vphantom{\begin{bmatrix} -1.9425 \\ -1.5621 \\ \vdots \\ 1.4335 \\ 1.4335 \end{bmatrix}} \right\} 44 \text{ rows}$$

2. Calculation of covariance matrix (S) of matrix X

$$S = \text{covariance matrix of } X = \begin{bmatrix} 1 & 0.3011 & 0.3335 & -0.4168 & 0.2604 & 0.6447 \\ 0.3011 & 1 & 0.0801 & -0.4742 & 0.2574 & 0.6494 \\ 0.3335 & 0.0801 & 1 & -0.0964 & 0.0249 & 0.2588 \\ -0.4168 & -0.4742 & -0.0964 & 1 & -0.9122 & -0.6115 \\ 0.2604 & 0.2574 & 0.0249 & -0.9122 & 1 & 0.3674 \\ 0.6447 & 0.6494 & 0.2588 & -0.6115 & 0.3674 & 1 \end{bmatrix}$$

Symmetrical -> **eigenvectors are orthogonal!**

Matlab command: S=cov(X)

Example: Interpretation of scores and loadings with semiconductor data (Continued)

3. Calculation of eigenvalues and eigenvectors of covariance matrix (S)

$P = \text{eigenvectors of matrix } S =$

$$\begin{bmatrix} -0.0127 & 0.4316 & 0.7058 & 0.0786 & 0.3936 & 0.3928 \\ -0.0803 & 0.4572 & -0.4518 & -0.6541 & 0.0501 & 0.3874 \\ -0.0002 & 0.0245 & -0.5264 & 0.5302 & 0.6422 & 0.1693 \\ -0.7551 & 0.0363 & 0.0697 & -0.2024 & 0.3560 & -0.5058 \\ -0.6253 & 0.1088 & -0.0702 & 0.4069 & -0.5065 & 0.4124 \\ -0.1791 & -0.7687 & 0.1041 & -0.2800 & 0.2164 & 0.4909 \end{bmatrix}$$


Loadings

$\Lambda = \text{eigenvalues of matrix } S =$

$$\begin{bmatrix} 0.0420 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.2204 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5865 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8700 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.2247 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3.0563 \end{bmatrix}$$

Matlab command: $[P, \Lambda] = \text{eig}(S)$

Example: Interpretation of scores and loadings with semiconductor data (Continued)

4. Calculation of scores matrix, T

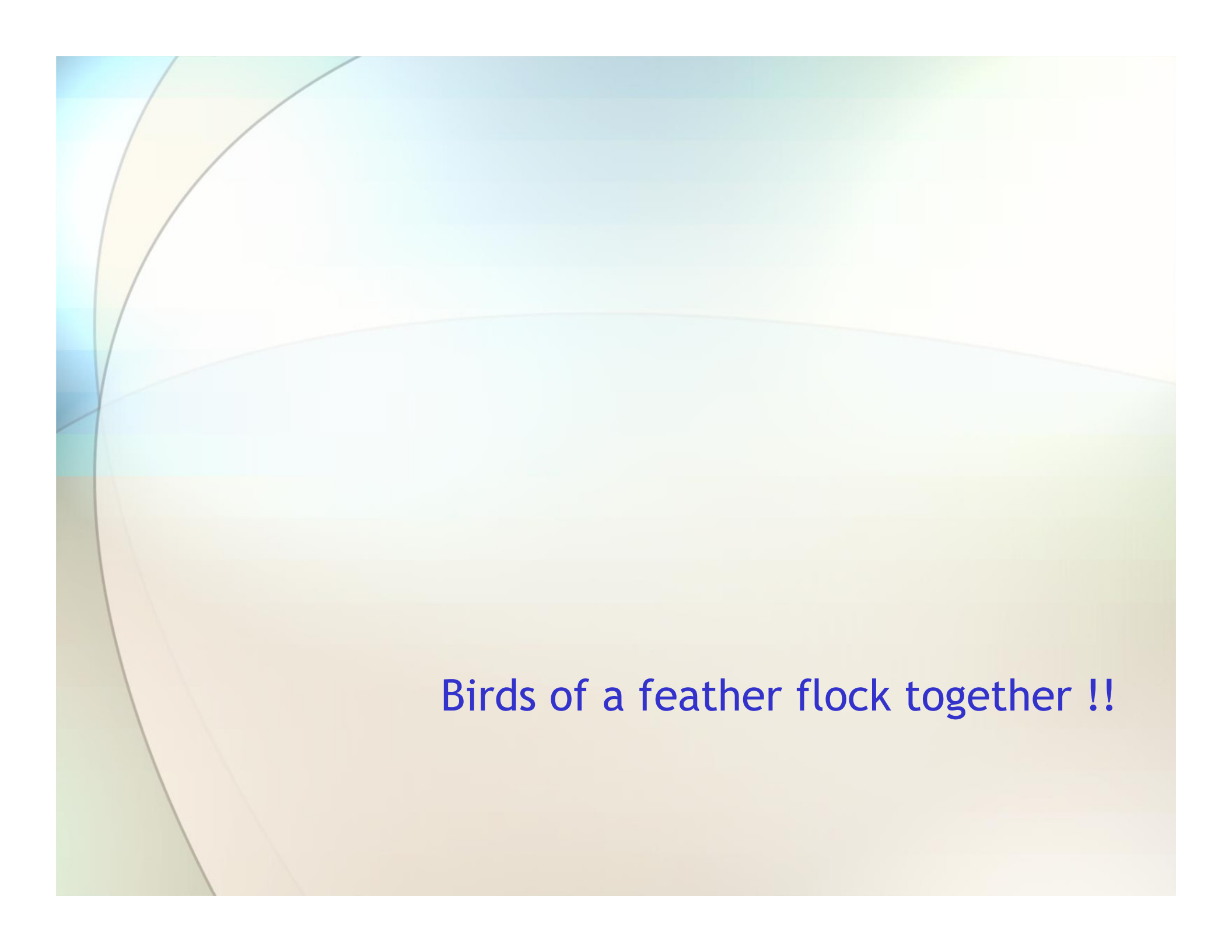
$$X = \text{auto-scaled matrix of } A = \begin{bmatrix} -1.9425 & -0.5401 & -0.5887 & 1.3495 & -0.9224 & -2.5417 \\ -1.5621 & 0.1188 & -0.5887 & 0.2139 & -0.2332 & -0.0565 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1.4335 & 0.1694 & -0.5887 & -0.6054 & 0.0789 & 0.7754 \\ 1.4335 & 0.5754 & -0.5887 & -1.5220 & 1.4833 & 0.7544 \end{bmatrix}$$
$$P = \text{eigenvectors of matrix } S = \begin{bmatrix} -0.0127 & 0.4316 & 0.7058 & 0.0786 & 0.3936 & 0.3928 \\ -0.0803 & 0.4572 & -0.4518 & -0.6541 & 0.0501 & 0.3874 \\ -0.0002 & 0.0245 & -0.5264 & 0.5302 & 0.6422 & 0.1693 \\ -0.7551 & 0.0363 & 0.0697 & -0.2024 & 0.3560 & -0.5058 \\ -0.6253 & 0.1088 & -0.0702 & 0.4069 & -0.5065 & 0.4124 \\ -0.1791 & -0.7687 & 0.1041 & -0.2800 & 0.2164 & 0.4909 \end{bmatrix}$$

$$\begin{aligned} \text{Size of score matrix, } T &= [X][P] \\ &= (44 \times 6) * (6 \times 6) \\ &= 44 \times 6 \end{aligned}$$

Matlab command: $T=X*P$

Example: Interpretation of scores and loadings with semiconductor data (Continued)

	PC6	PC5	PC4	PC3	PC2	PC1
$T = \text{Scores} =$	0.0812	0.8026	-0.9228	-0.0484	-0.7721	-3.3827
	0.0048	-0.6085	-0.8208	-0.6350	-0.8050	-0.8993
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	0.2372	0.0723	1.2781	-0.3728	0.1069	1.2485
	0.0224	0.3937	0.9300	0.1245	-0.9149	2.4383
	-0.0127	0.4316	0.7058	0.0786	0.3936	0.3928
	-0.0803	0.4572	-0.4518	-0.6541	0.0501	0.3874
$P = \text{Loadings} =$	-0.0002	0.0245	-0.5264	0.5302	0.6422	0.1693
	-0.7551	0.0363	0.0697	-0.2024	0.3560	-0.5058
	-0.6253	0.1088	-0.0702	0.4069	-0.5065	0.4124
	-0.1791	-0.7687	0.1041	-0.2800	0.2164	0.4909

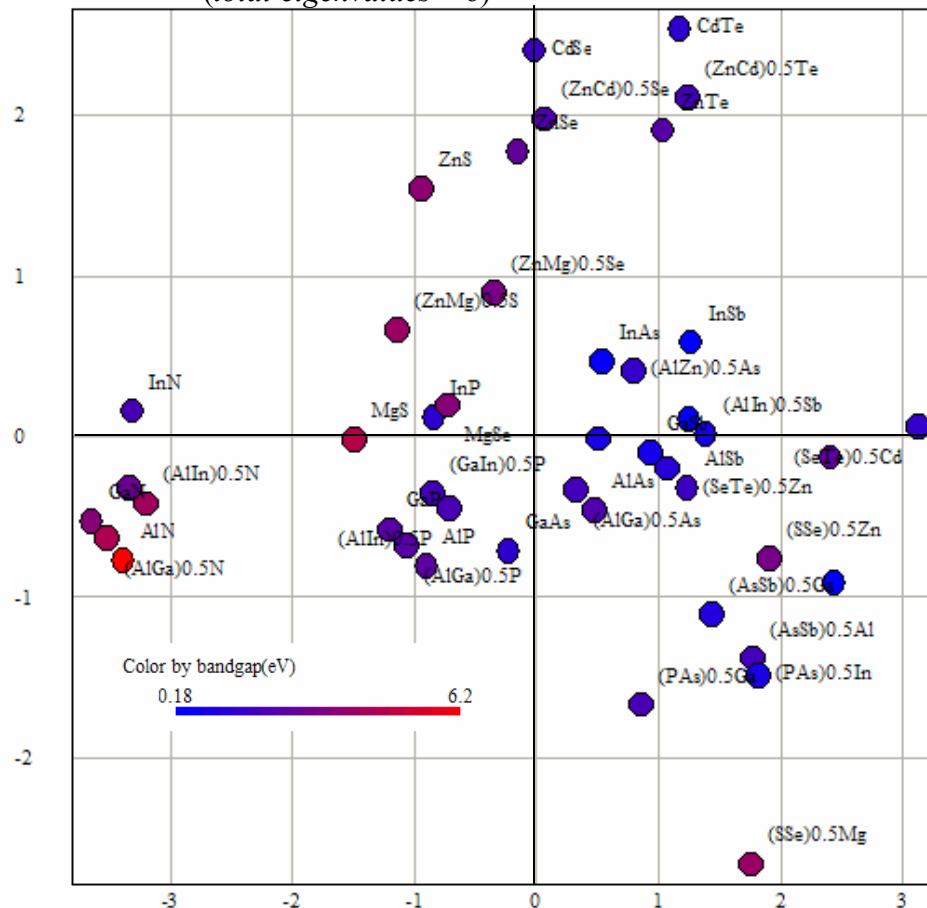


Birds of a feather flock together !!

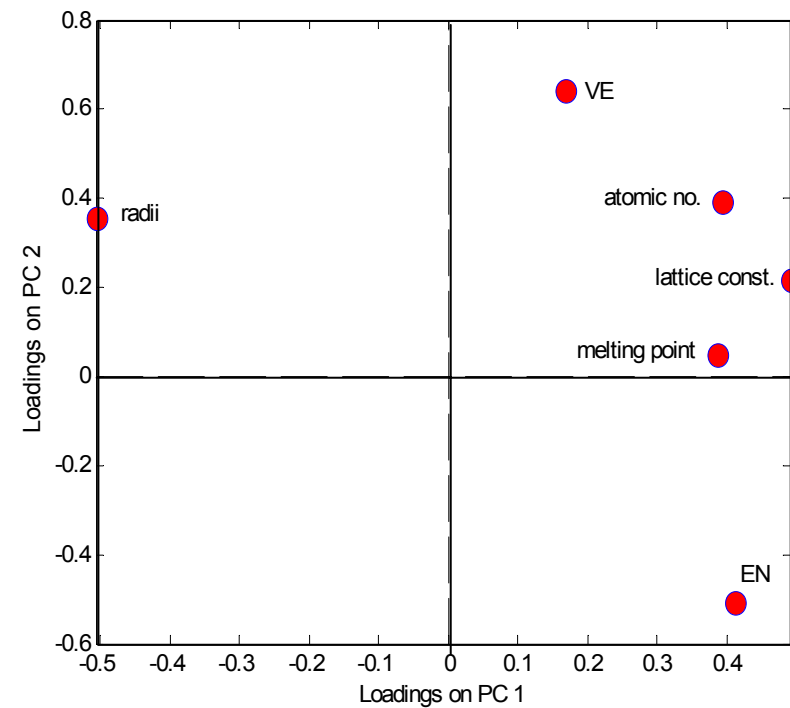
Example: Interpretation of scores and loadings with semiconductor data (Continued)

Score Plot

$$PC2(20.41\%) = \frac{1.2247}{(total\ eigenvalues = 6)} \times 100 = 20.41\%$$



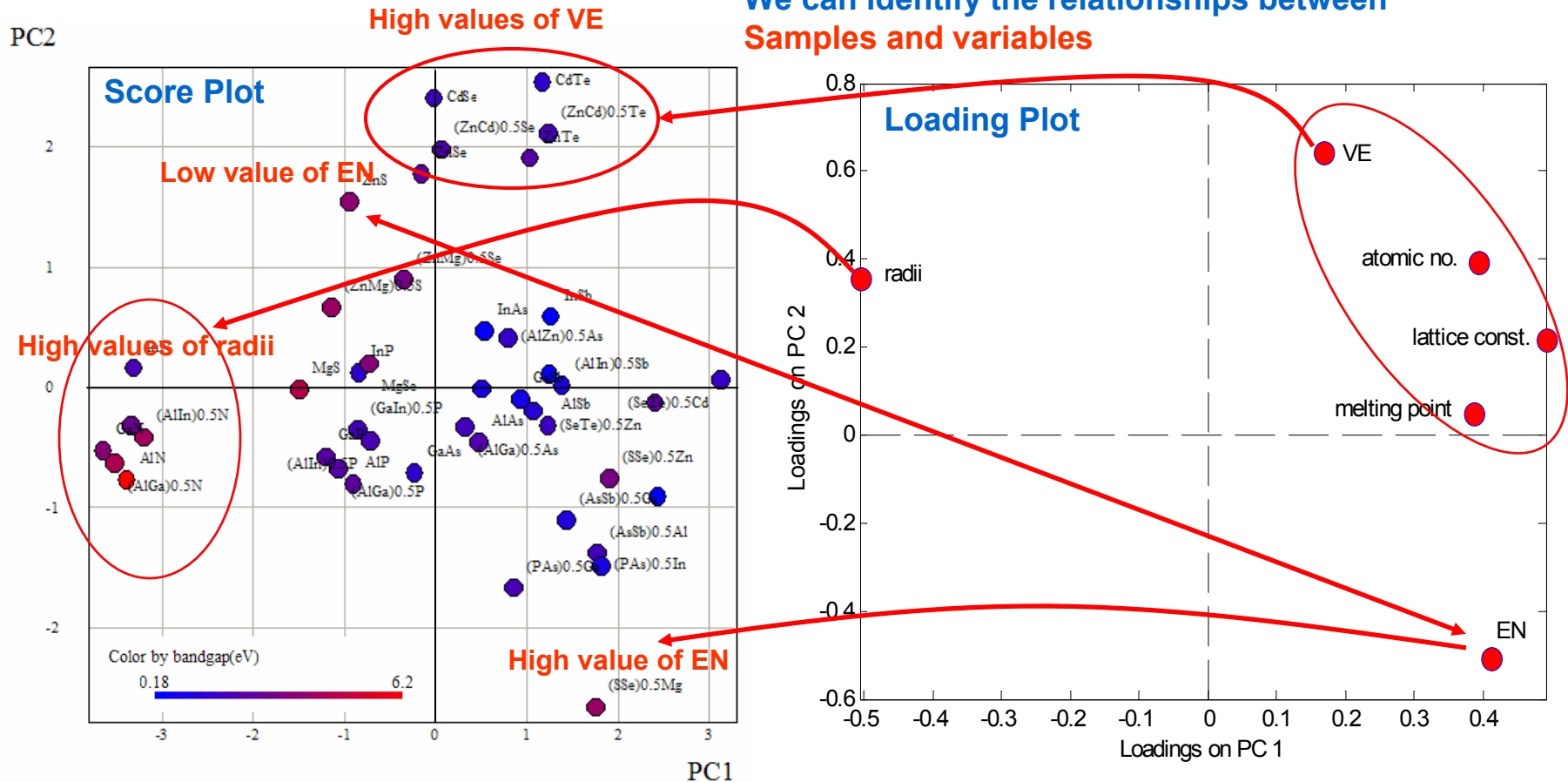
Loading Plot



$$PC1(50.94\%) = \frac{3.0563}{(total\ eigenvalues = 6)} \times 100 = 50.94\%$$

Example: Interpretation of scores and loadings with semiconductor data (Continued)

By comparing the score and loading plot, We can identify the relationships between Samples and variables



On the score plot,
“Sit together”: similar behavior between samples
 ex.) Nitrides

On the loading plot,
“Sit together”: similar behavior between variables
 ex.) VE, atomic no. lattice const., and melting point